Agricultural Mobile Apps in Sri Lanka: A Systematic Review

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Abstract— **Digital technologies are increasingly recognized as essential tools for enhancing smallholder productivity and agricultural resilience. However, in Sri Lanka, the development and deployment of such technologies remain fragmented, raising questions about whether existing digital platforms adequately support paddy farmers and agricultural officers. This systematic review examines the landscape of mobile- and web-based agricultural solutions in Sri Lanka and assesses their functional coverage. Evidence from Sri Lanka-specific documents—including peer-reviewed articles, theses, government publications, project reports, and official app or platform descriptions published between 2000 and 2025—was synthesised. Additional international literature is cited for methodological guidance and contextual comparison but was not included in the core evidence set. Searches were conducted across major academic databases, Sri Lankan government and international agency websites, and the Google Play and Apple app stores, following PRISMA 2020 guidelines. Data were extracted on platform developers, target users, delivery channels, and functions. The review reveals a dispersed ecosystem of government, private-sector, and hybrid platforms, each addressing only a subset of farmer needs. Weather and advisory information are available through separate services, static technical content is abundant on institutional portals, and some operator-led platforms provide personalized advice or market information. However, farm-level data management, integrated resource requests, and structured two-way digital communication with agricultural officers are largely absent. No comprehensive, integrated national-level platform was identified that supports the full spectrum of paddy farmers' information and service requirements. These findings highlight the need for coordinated digital agriculture strategies, interoperable system design, and supportive policy frameworks to move towards an integrated, farmer-centered platform ecosystem in Sri Lanka**.

***Keywords*—Digital agriculture, Mobile applications, Sri Lanka, Adoption barriers, Communication, agriculture automation**

I. Introduction

Automation in agriculture comprises a spectrum of technologies that enable farming operations to become more data-driven, timely, and efficient[1], [2]. In smallholder settings, this automation is often informational rather than purely mechanical: mobile applications, web platforms, and ICT-based decision support tools help farmers plan, monitor, and manage activities by delivering targeted information, recommendations, and alerts. When appropriately designed, these systems can support farmers in making decisions on crop choice, input use, irrigation, pest and disease management, and marketing, thereby reducing uncertainty and dependence on sporadic face-to-face advice.

At the global level, digitalization has become a central in strategies to modernize agriculture and rural areas. International assessments describe how ICTs, mobile services, and online platforms can connect smallholders to knowledge, networks, and institutions that were previously out of reach [3], [4], [5],[6],[7]. These tools include SMS-based advisory services, smartphone apps, and farm information systems that disseminate weather forecasts, agronomic guidance, and market prices directly to farmers [6]. Empirical studies show that timely access to such information can improve farmers' knowledge and, in some contexts, influence technology adoption, bargaining power, and incomes, though impacts are heterogeneous and strongly context-specific [8],[9], [10], [11]

Beyond basic advisory services, the concept of "digital agriculture" or "smart farming" reflects a broader integration of ICTs, connectivity, data analytics, and automation into agrifood systems. Smart farming frameworks emphasis the use of sensors, remote sensing, big data, and interconnected platforms to monitor crops and resources, support complex decision-making, and partially automate routine tasks[12], [13] . While much of this agenda has developed in high-income or commercial farming systems, elements of smart farming—such as decision support, data capture, and real-time alerts—are increasingly being adapted for smallholders in low- and middle-income countries[14]. In these settings, even relatively simple mobile or web-based platforms can play an "automation" role by structuring and scheduling flows of information that were previously fragmented or delayed.

Sri Lanka reflects many of these global dynamics. Agriculture's share in national GDP has declined, yet the sector remains vital for food security, rural employment, and poverty reduction. Rice is the staple food, and paddy cultivation dominates irrigated and rainfed lowland systems. Paddy farmers manage production under monsoon-driven rainfall, increasing climate variability, and recurrent pest and disease outbreaks, while facing constraints related to input access, price volatility, and uneven coverage of public extension services. Recognizing these challenges, Sri Lanka has articulated policy directions that explicitly highlight the role of ICTs and digital tools in agricultural transformation, including the National e-Agriculture Strategy, the National Agricultural Policy, and the National Digital Policy[15][16]. Development partners, notably the World Bank through the Agriculture Sector Modernization Project and the earlier e-Sri Lanka Development Project, have also emphasized ICT components within broader sector modernization and digitalization efforts [17], [18], [19].

Aligned with these strategies, multiple digital agriculture initiatives and platforms have emerged in Sri Lanka over the past decade. Examples include mobile operator–led services such as Govi Mithuru, which delivers crop-stage–based

advisory messages via IVR and SMS to registered farmers[20]; market information and trading platforms such as Dialog Tradenet[21]; web-based knowledge and advisory portals maintained by the Department of Agriculture; agrometeorological bulletins and seasonal forecasts disseminated by the Department of Meteorology[22], [23], [24]; and Sri Lanka-specific rice knowledge resources hosted by the International Rice Research Institute. Collectively, these systems provide agronomic recommendations, seasonal guidance, static pest and disease information, market prices, and general weather-related advice.

However, the existing digital agriculture landscape appears to be fragmented across institutions, technologies, and functions. Different platforms are typically developed, owned, and maintained by separate government agencies, private companies, or donor-funded projects, often with limited interoperability or coordination. Many services focus on a narrow subset of farmer needs—for example, weather information, market prices, or generic cultivation tips—rather than offering a comprehensive, integrated service environment. From the perspective of paddy farmers and agricultural officers, it is not evident that there is a single, central platform through which users can simultaneously obtain real-time weather updates, targeted pest and disease alerts, tools for farm-level data recording, mechanisms to request inputs and services, structured knowledge and training content, and robust two-way communication channels.

Despite a substantial global literature on ICTs for agriculture and mobile-based advisory services[19], Sri Lanka-specific evidence remains scattered across peer-reviewed articles, government and project reports, and app-store descriptions. Existing accounts typically focus on individual initiatives or policies, without systematically mapping and comparing the full range of mobile applications and web platforms relevant to Sri Lankan agriculture. As a result, policymakers, system designers, and researchers lack a consolidated overview of what platforms exist or have existed, which actors are involved, what functions they deliver in practice, which systems are active or discontinued, and where key functional and institutional gaps lie.

To address this gap, the present study conducts a systematic review of mobile applications, web platforms, and related digital services for agriculture in Sri Lanka, with particular attention to their relevance for paddy farmers and agricultural officers. Following contemporary guidance on systematic review methodology[25], [26], [27], and drawing on both peer-reviewed and grey literature as well as formal app and platform documentation, this review seeks to answer the following Research Questions (RQs):

RQ1: What agricultural mobile applications, web platforms, and related digital services have been developed or deployed in Sri Lanka, and who are their primary developers and target users?

RQ2: What core functions do these platforms provide, particularly with respect to real-time weather information, pest and disease alerts, farm data recording, resource and service requests, knowledge and extension content, and two-way communication between farmers and agricultural officers?

RQ3: How are these platforms distributed across government, private-sector, and hybrid arrangements, and which are currently active, inactive, or discontinued?

RQ4: What empirical evidence exists on the adoption, use, and effects of these platforms on farmers' knowledge, decision-making, and agricultural outcomes?

RQ5: To what extent do existing platforms collectively or individually constitute a central, integrated digital agriculture solution for paddy farmers in Sri Lanka, and what are the main gaps that future system design and policy should address?

II. METHODOLOGY

This systematic review was done according to the PRISMA 2020 statement (1), followed an advance protocol specifying eligibility criteria, information sources, search strategies, selection procedures, data extraction, and synthesis methods.

The study was conducted as a systematic review of peer-reviewed literature, grey literature, and formal documentation of mobile and web-based digital agriculture platforms described for use in Sri Lanka. Only Sri Lanka-specific documents that met the eligibility criteria were included in the main synthesis, while international studies were used solely for background and methodological orientation. The review combined elements of traditional systematic reviews, but to align with the systematic principles, comprehensive searching, inclusion/exclusion criteria definition, screening, structured data extraction, and transparent reporting was done.
The primary objective was to identify and characterise agricultural mobile applications and web platforms described for use in Sri Lanka, and to assess the extent to which their functions collectively or individually constitute an integrated, central platform providing: (i) real-time weather information, (ii) pest and disease alerts farm, (iii) data recording, (iv) resource or service requests from the officers, (v) knowledge and advisory content, and (vi) two-way communication between agricultural officers and paddy farmers.

A. *Eligibility criteria*

Eligibility criteria were defined using PICo(Population, Intervention, Context) structure and are summarised in Table1

TABLE 1 ELIGIBILITY CRITERIA (PICO FRAMEWORK)

Dimension	Inclusion	Exclusion
Population/ Context	Agriculture in Sri Lanka (national or sub-national; including paddy farmers)	Non-Sri Lanka contexts; non-agricultural domains (health, education, etc., without agri)
Intervention	Digital agriculture platforms: mobile apps, web portals, structured SMS/USSD/IVR services	Purely mechanical tools; ad hoc SMS campaigns; generic ICT projects with no agri component
Purpose/ Content	Agronomic advice, weather, pests/diseases, farm records, markets, communication, knowledge	ICT systems unrelated to farming decisions or services
Document type	Peer-reviewed, government/NGO reports, project docs, app store & official web descriptions	Opinion pieces, editorials, unverifiable internal documents
Time frame	2000–2025	Pre-2000
Language	English (with selective use of Sinhala/Tamil from official sites)	Other languages not practically translatable

B. Search Strategy

The search strategy incorporated academic databases, grey literature sources, and app store or platform websites, aligning with recommended approaches for minimising publication bias in systematic reviews.

Academic databases were searched between 2000 and 2025, including Scopus, the Web of Science Core Collection, IEEE Xplore, and Google Scholar, the latter of which was used to ensure supplementary coverage. Grey literature searches systematically targeted major Sri Lankan government websites—such as the Ministry of Agriculture, the Department of Agriculture (DoA), the Agrarian Development Department, the Information and Communication Technology Agency (ICTA), and the Department of Meteorology—as well as key international and regional organisations. These included the Food and Agriculture Organization (FAO), World Bank, and other initiatives relevant to ICT for agriculture. Site-restricted Google queries (e.g., site:gov.lk agriculture mobile app, site:fao.org "Sri Lanka" agriculture ICT) were used to locate relevant grey literature efficiently.

Searches of the Google Play Store and Apple App Store were also conducted using predefined English keywords such as "Sri Lanka agriculture" and "paddy farmer Sri Lanka," with Sinhala and Tamil equivalents applied where feasible. App titles and brief descriptions were screened initially, and potentially relevant applications were assessed further through detailed descriptions and linked developer or organisational websites.

The Boolean search string used for which adapt each datbase queries was formulated

("Sri Lanka" OR "Sri Lankan") AND (agricultur OR farm* OR paddy OR "rice farming") AND (app OR "mobile application" OR smartphone OR "web platform" OR "information system" OR "decision support system" OR "digital agriculture" OR "e-agriculture" OR ICT))*

C. Study Selection

The eligibility criteria in Table 1 were then applied to exclude non-Sri Lankan empirical studies from the core evidence set, although some of these studies are still cited in the introduction and discussion to contextualize the Sri Lankan findings.

Because this review was conducted by a single researcher, screening was performed by one reviewer, following the protocol. This is a deviation from ideal dual-screening practice recommended in some guidelines [26] and is noted as a limitation.

The selection process involved two stages

1. Title and Abstract/Summary Screening—Titles and abstracts (for academic items), executive summaries (for reports), and short descriptions (for app entries) were screened against the eligibility criteria.

Items clearly not meeting inclusion criteria were excluded. Items that appeared potentially relevant progressed to full-text/description review.

2. Full-Text/ Full-Description Screening—Full texts of publications and reports were retrieved where possible. For apps and platforms, complete app-store descriptions and associated official web pages were examined.

Each item was assessed against inclusion/exclusion criteria.

For excluded full-texts, the primary reason for exclusion (e.g., not Sri Lanka-specific, no agricultural component, no digital system) was recorded.

The overall selection process is summarized in a PRISMA-style flow diagram, Fig. 1, [25]

D. Data Extraction

A structured data-extraction form was designed on a subset of items. Extraction was conducted by the single reviewer in a systematic, platform-centered manner: where multiple sources described the same platform, information was merged into one consolidated entry. When specific data items were not available, they were recorded as "not reported" rather than assumed absent. By usinf a structured set of main data-extraction fields to ensure consistency and completeness across all included platforms as shown in table 2

TABLE 2 MAIN DATA EXTRACTION FIELDS

Category	Data items
Source details	Author/organisation, year, country, document type (article, report, app listing, website)
Platform ID	Platform/app name; alternative names; version (if relevant)
Developer	Organisation(s); developer type (government, private, NGO, hybrid)
Target users	Paddy farmers; other crop farmers; livestock; extension officers; mixed stakeholders
Geographic coverage	National; specific provinces/districts; pilot sites
Platform type	Android app; iOS app; web platform; SMS/USSD; IVR; multi-channel
Languages	Sinhala; Tamil; English; others
Functional features	Weather info; pest/disease info/alerts; farm data recording; resource/service requests; market info; knowledge/extension content; two-way communication; decision support tools
Technical aspects	Offline capability; data storage (local/cloud); integration with other systems (e.g., farmer registry, met data)
Status & scale	Active/inactive; last update; reported users/downloads; scale of deployment
Evaluation & outcomes	Study design (if any); adoption/use; impacts on knowledge, practice, yields, income; key findings
Funding/affiliations	Funding source; potential conflicts of interest

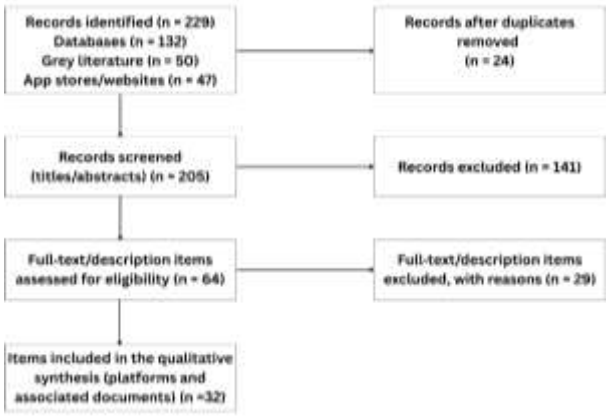

Fig 1 *PRISMA-style flow diagram*

(*Compiled by the author*)

III. RESULTS AND DISCUSSION

A. Functional coverage of key platforms

To assess functional coverage, a compact feature matrix was constructed for the main, well-documented platforms (Table 3). No platform combined all seven functional domains in a single, fully integrated environment for paddy farmers and agricultural officers.

Across the documents, the review identified several dozen references to digital initiatives, which, when duplicates and concept-only proposals were merged, corresponded to around 15–20 distinct operational or pilot platforms. These can be broadly grouped by developer and platform type (Table 4).

TABLE 3 FUNCTIONAL FEATURE MATRIX FOR SELECTED PLATFORMS

Platform / Service	Weather	Pest/disease	Farm records	Resource requests	Market information	Knowledge	Two-way communication
Govi Mithuru (IVR/SMS advisory)[20]	P	Y	N	N	P	Y	P
Dialog Tradenet (market info/trading)[21]	N	N	N	P	Y	P	N
DoA online portals (crop info, manuals)	N	static	N	P	N	Y	N
Dept. of Meteorology agromet services	Y	P	N	N	N	Y	N
IRRI Rice Knowledge Bank – Sri Lanka section[28]	N	S	N	N	N	Y	N
Nat. e-Agriculture Strategy pages/meta-portal	N	N	N	N	N	P	N

(*Compiled by the author*)

TABLE 4 OVERVIEW OF PLATFORMS BY DEVELOPER AND CHANNEL

Category	Main sub-categories	Approx. count
Developer type	Government (DoA, Dept. of Meteorology, etc.)	7
	Private sector (e.g., Dialog Axiata)	4
	Hybrid gov–private / gov–NGO	4
	International portals with Sri Lanka content	2
Delivery channel	Web portals / websites	6
	SMS/USSD and IVR services	4
	Smartphone apps (Android/iOS)	2
	Multi-channel (web + mobile + SMS/IVR)	3

(*Compiled by the author*)

B. Summary of main findings

This review found a diverse but fragmented set of digital agriculture in iteratives in Sri Lanka. A variety of government, private-sector, hybrid, and international platforms provide selected services—such as advisory content, market information, and weather bulletins—but these are spread across institutions and channels and typically address only a subset of paddy farmers' needs.

Key gaps include—Limited integration of core functions—weather, pests/diseases, records, resource requests, knowledge, and communication, in any single platform, Weak support for farm-level data management and digital resource/service requests, Minimal structured two-way digital communication between paddy farmers and agricultural officers. Despite policy aspirations and multiple initiatives, there is no documented, comprehensive, integrated national-level platform that supports the full spectrum of information and service requirements for paddy farmers.

A. Comparison with international evidence

The studies cited in this subsection are primarily non-Sri Lankan and were not part of the 53 included documents. They are drawn from the broader ICT4Ag and digital agriculture literature to situate the Sri Lankan findings in an international context.

The Sri Lankan experience is consistent with global ICT4Ag and digital agriculture literature:

Many services internationally focus on single-function information delivery—for example, price information or agronomic SMS advice—rather than fully integrated decision-support systems[6], [10], [29]

Empirical studies indicate that such services can improve farmers' knowledge and, in some cases, affect technology adoption and marketing behaviour, but measured impacts on yields and incomes are mixed and context-dependent ([7], [8], [10].

Reviews of ICTs and extension emphasise the difficulty of moving from pilot projects to sustained, scaled, and integrated[30], [31].

From a smart farming and Agriculture 4.0 perspective, Sri Lankan platforms largely operate at the "digital advisory" level—disseminating information and basic recommendations—rather than forming part of sensor-driven, data-intensive farm management architectures [12], [13], [14] .This confirms that Sri Lanka is in an early

phase of digitalization, with ample scope for more integrated and data-driven solutions.

B. Barriers and Challenges in the Sri Lankan Context

Thematic analysis across documents suggests several interconnected challenges that help explain the observed fragmentation:

Institutional fragmentation—Separate mandates and systems across the Department of Agriculture, Department of Meteorology, Agrarian Development, ICTA, and private operators make joint planning and integration difficult.

Technical fragmentation—Absence of shared farmer identifiers, common data standards, and open APIs limits interoperability between platforms, preventing data and services from being combined into a single user experience.

Socio-economic and infrastructural constraints—Unequal access to smartphones, reliable internet, and digital literacy—especially among smallholders in remote or poorer areas—constrains adoption, echoing patterns documented in other low- and middle-income countries[6], [20], [32].

Sustainability and business models—Several platforms appear project-based or donor-funded, with uncertain long-term funding and maintenance, like global experiences where many digital services fail to scale or sustain beyond initial funding cycles [9].

These barriers collectively contribute to a situation where many components exist, but remain unconnected, under-utilized, or short-lived.

C. Research Gaps and Future Directions

The review identifies several priority areas for further research and innovation in Sri Lanka's digital agriculture:

Design and evaluation of integrated platforms—There is a need to design, pilot, and rigorously evaluate platforms that deliberately integrate weather, pest/disease alerts, record-keeping, resource requests, knowledge content, and two-way communication in a single environment tailored for paddy farmers.

User-centered and inclusive design studies—More work is needed on interface design, language, modality (voice/text), and accessibility to ensure that platforms are usable for farmers with varying literacy, age, and technology access [6], [33]. These constraints mirror regional patterns in mobile access and usage highlighted in Asia-Pacific reports[32], [33], [34].

Impact and cost-effectiveness research—Strong longitudinal or experimental studies could clarify which combinations of functions and delivery models yield the greatest benefits in terms of knowledge, resilience, productivity, and incomes, and at what cost.

Governance, interoperability, and data-sharing Research on institutional arrangements, incentives, and regulatory frameworks can support development of interoperable platforms that cross institutional boundaries while addressing data privacy and security. International organizations emphasize that realising the benefits of digital agriculture requires coherent policy frameworks, data governance, and cross-sector coordination [33], [35].

IV. CONCLUSION

This systematic review synthesized evidence on mobile and web-based agricultural platforms in Sri Lanka, with a focus on paddy farmers and agricultural officers. The analysis of 53 Sri Lanka-specific documents shows that, while multiple government, private, hybrid, and international initiatives exist, they form a fragmented ecosystem of single- or limited-function services. No comprehensive, integrated national-level platform was identified that supports the full spectrum of information and service needs of paddy farmers and agricultural officers. International studies cited in this paper reinforce that such fragmentation and limited integration are common challenges in digital agriculture globally.

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